Journal Target Recommendation: '02' DGP Architecture Comparison

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Purpose. This note recommends target journals for the DGP architecture comparison manuscript (then numbered '02', now paper 01, `analysis/report/01-dgp-mean-moderation-vs-mvn/`). It was generated on 2026-04-08 from the Zotero nof1 collection (50 items) and a reading of the manuscript draft of that date (`docs/02-dgp-mean-moderation-vs-mvn.tex`, which is no longer in the repository; only an annotated PDF of that draft remains in `docs/`). It was reviewed on 2026-10-07 for notation, reference accuracy and project conventions.

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Notation index and glossary

This note uses no mathematical symbols. Architecture names follow `analysis/report/NOTATION.md`, Part 3.

Table 1: Terms used in this note, with meaning and status

Term	Meaning	Status
mean-moderation architecture	the biomarker scales the treatment effect in the conditional mean (formerly Architecture A; data value `mean_moderation`)	canonical
covariance-moderation architecture	the biomarker-response correlation differs by treatment state within a multivariate normal (MVN) draw (formerly Architecture B; data value `mvn`)	canonical
DGP	data-generating process	canonical
Path B	the strengthened revision plan for the manuscript as of 2026-04-08 (1,000 or more replicates, evaluated alternative analysis strategies, power curves)	local
collection precedent	a paper in the Zotero nof1 collection published in the named journal	local

Glossary. *N-of-1 trial*: a multiple-crossover trial in a single participant; *aggregated N-of-1 trial*: many such series pooled in one mixed model. *Carryover*: persistence of a drug effect after discontinuation.

Manuscript summary

Note (2026-10-07). This section describes the manuscript as it stood on 2026-04-08. The power-loss figures below are superseded and should not be quoted. When the comparison was re-run at a matched total of $N = 70$

for all designs (2026-08-08), the Hybrid-design carryover loss was 6.9 percent under mean moderation and 15.4 percent under covariance moderation (`analysis/report/NOTATION.md`, Part 2). Paper 01 has since been re-scoped as a refined data-generating process following Hendrickson et al. (2020): the published step coupling makes the coupling gap zero, so the interaction vanishes at any positive half-life, and graded coupling ($2^{-t_{sd}/t_{1/2}}$ off drug, 0 before exposure) removes the collapse (docs/37); under graded coupling, covariance-based binding and mean moderation that fades with the drug effect lose power almost identically (paper 01, abstract). The journal reasoning below should be revisited against the current manuscript before it is acted on.

'Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover' formalizes the distinction between direct mean moderation (the mean-moderation architecture) and differential correlation within an MVN draw (the covariance-moderation architecture) as data-generating processes for biomarker-treatment interaction simulations. The central finding, as stated in the 2026-04-08 draft, was that the choice of DGP architecture determines whether carryover effects reduce statistical power modestly (~10-15%, mean moderation) or substantially (~40-60%, covariance moderation). The manuscript provides mathematical formalization, biological rationale for each architecture, simulation evidence from N-of-1 trial designs, and guidance for DGP selection.

Methodological positioning

The manuscript sits at the intersection of three domains:

- Statistical simulation methodology (DGP specification and its consequences)
- N-of-1 and crossover trial design (carryover, power analysis)
- Biomarker-treatment interaction modeling (predictive biomarker validation)

Most similar papers in the Zotero nof1 collection

Table 2: Most similar papers in the Zotero nof1 collection with similarity and journal

Paper	Similarity	Journal
Shi & Ye (2024): behavioral carryover and power in crossover trials	Highest	*Biometrics*

Paper	Similarity	Journal
Blackston et al. (2019): simulation comparison, aggregated N-of-1 vs. RCT	High	*Healthcare*
Wang & Schork (2019): power and design in crossover N-of-1	High	*Healthcare*
Senn (2016): variance components and personalized medicine	High	*Statistics in Medicine*
Schork (2022): serial correlation in personalized studies	High	*Harvard Data Science Review*
Haller & Ulm (2018): biomarker-treatment interaction estimation	High	*Trials*
Senn (2019): sample size for N-of-1 (see note)	Moderate	*Statistical Methods in Medical Research*
Tang & Landes (2020): t-tests for N-of-1 with serial correlation	Moderate	*PLoS ONE*
Hawksworth et al. (2024): methodological review of randomized N-of-1 trials	Moderate	*Trials*

Attribution corrections (2026-10-07). Six citations in the original table and in the ranked list below were misattributed or could not be found when checked against PubMed, Crossref and the Zotero library. Three of them match Zotero attachment file names rather than the item metadata, which is the likely source of the error. "Yap, Wang & Harhay (2024)" in *Biometrics* could not be found; the collection's 2024 *Biometrics* paper on carryover and power in crossover trials is Shi and Ye (2024), and the entry is taken to be that paper. "Chen et al. (2024)" is Hawksworth et al. (2024). "Senn & Julious (2024)" is Senn (2024), a single-author paper. "Mitchell et al. (2024)" is Chatters et al. (2024). "Arends et al. (2019)" in *Statistical Methods in Medical Research* could not be found; the collection's 2019 paper on N-of-1 sample size in that journal is Senn (2019), and the entry is taken to be that paper (an inferred identification). "Liu et al. (2021)" in *Contemporary Clinical Trials Communications* could not be found in PubMed or in the Zotero library and is flagged as unverified.

Ranked journal targets

Rank 1: *Statistics in Medicine*

- **Collection precedent:** Senn (2016)
- **Rationale:** Premier journal for statistical methodology in clinical trial design. The core contribution (in the 2026-04-08 draft, that DGP architecture choice produces 40-60% divergent power estimates under carryover; superseded, see the note under the manuscript summary) is a methodological finding with implications for how the field designs and reports simulation studies. The journal's readership (biostatisticians designing trials) is the primary audience who would act on this finding. The Path B strengthened version (1,000+ replicates, evaluated alternative analysis strategies, power curves) meets the journal's expectations for rigor.
- **Risk:** Competitive acceptance rate. Must demonstrate the finding generalizes beyond the specific pmsimstats parameterization.

Rank 2: *Biometrics*

- **Collection precedent:** Shi & Ye (2024)
- **Rationale:** Shi & Ye (2024) is the closest comparator paper in the collection: carryover's impact on power in crossover designs, published in *Biometrics* in 2024. The '02' manuscript extends this by showing that the mechanism by which the biomarker-treatment interaction is generated determines the magnitude of power loss. Natural companion to Shi & Ye.
- **Risk:** Expects stronger theoretical results (formal proofs, asymptotic analysis). May view the scope as too narrow for a generalist biometrics readership. Would need broader framing in terms of covariance-vs-mean signal detection.

Rank 3: *Trials*

- **Collection precedent:** Haller & Ulm (2018), Hawksworth et al. (2024), Senn (2024), Chatters et al. (2024)
- **Rationale:** Highest concentration of N-of-1 methodology papers in the collection (4 of 50). Haller & Ulm (2018) is essentially the same methodological genre as '02' (simulation study of biomarker-treatment interaction estimation). Open access, practical orientation, readership includes both methodologists and trialists.
- **Risk:** Lower impact factor. Theoretical depth may exceed the journal's typical level, though this is minor.

Rank 4: *Statistical Methods in Medical Research*

- **Collection precedent:** Senn (2019) (listed originally as "Arends et al. (2019)"; see the attribution note)
- **Rationale:** Publishes simulation-based methodological work for clinical trial design. Senn (2019) is the same genre (power/sample size for N-of-1). Occupies middle ground between theoretical focus of *Biometrics* and practical focus of *Trials*. Accepts longer methodological papers with extensive simulation.
- **Risk:** Slower review times. Less visibility in the N-of-1 community.

Rank 5: *Contemporary Clinical Trials Communications*

- **Collection precedent:** Sturdevant & Lumley (2021); Liu et al. (2021) (unverified: not found in PubMed or in the Zotero library)
- **Rationale:** Relevant work on carryover testing (Sturdevant & Lumley) and, if the second entry exists, crossover design. Good fallback. Practical framing and clinical application (prazosin-PTSD) fit well.
- **Risk:** Lower visibility and impact than journals above.

Rank 6: *Harvard Data Science Review*

- **Collection precedent:** Schork (2022)
- **Rationale:** Schork (2022) on related statistical machinery was published here. HDSR publishes methodological perspectives.
- **Risk:** More of a review/perspective venue. Heavy simulation results in the Path B version may not fit format expectations.

Rank 7: *Healthcare* (MDPI)

- **Collection precedent:** Blackston et al. (2019), Wang & Schork (2019)
- **Rationale:** Two of the most similar simulation-comparison papers were published here. Fast review, open access.
- **Risk:** Lower standing of MDPI journals among methodologists. If strengthened to Path B standards, the manuscript would suit a higher-impact venue.

Primary recommendation

The rankings in this note are editorial judgments (argued), based on collection precedent and fit; no acceptance rates, impact factors or review times were looked up.

Target: *Statistics in Medicine*. The manuscript's central claim is about simulation methodology itself. *Statistics in Medicine* reaches the audience most likely to change their practice.

Backup: *Trials*. If *Statistics in Medicine* considers the scope too specialized, *Trials* has the most active N-of-1 methodology pipeline and strong fit with collection precedents.

References

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